A. The proof of the non-submodularity of $\sigma_\star^2(\mathcal{A})$ for some $\mathbf{\Sigma}$

Proof: Assume for two subsets $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{N}$ and an element $c \in \mathcal{N} \backslash \mathcal{B}$, we have

$$\mathbf{\Sigma}_{\mathcal{A}\mathcal{A}} = [1], \mathbf{\Sigma}_{\mathcal{B}\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{\Sigma}_{\{\mathcal{A},c\}\{\mathcal{A},c\}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{\Sigma}_{\{\mathcal{B},c\}\{\mathcal{B},c\}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then $\sigma_\star^2(\mathcal{A}) = 1$, $\sigma_\star^2(\mathcal{B}) = \frac{1}{2}$, $\sigma_\star^2(\{\mathcal{A}, c\}) = \frac{1}{2}$, and $\sigma_\star^2(\{\mathcal{B}, c\}) = \frac{1}{2}$, which means $\sigma_\star^2(\{\mathcal{A}, c\}) - \sigma_\star^2(\mathcal{A}) \leq \left(\sigma_\star^2(\{\mathcal{B}, c\}) - \sigma_\star^2(\mathcal{B})\right)$. Therefore, there exists at least one $\mathbf{\Sigma}$ such that the resulting risk control set function $\sigma_\star^2(\mathcal{K})$ is not submodular. ■

B. Proof of Eq. (3)

Proof: Since the covariance matrix $\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}$ is semi-definite, we can do eigendecomposition of $\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}$ as $\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where $\mathbf{V}\mathbf{V}^T = \mathbf{I}$ and $\mathbf{\Lambda} = \mathsf{Diag}(\lambda_1, \ldots, \lambda_K)$, with λ_1 be the largest eigenvalue of $\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}$. Then we have

$$\sigma_\star^2(\mathcal{K})^2 = \frac{1}{\mathbf{1}_K^T \mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}^{-1} \mathbf{1}_K} = \frac{1}{\mathbf{1}_K^T \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T \mathbf{1}_K} \leq \frac{\lambda_1}{\mathbf{1}_K^T \mathbf{V}\mathbf{I}\mathbf{V}^T \mathbf{1}_K} \leq \frac{\lambda_1}{\mathbf{1}_K^T \mathbf{1}_K} \leq \lambda_1 = \|\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}\|_2 \leq \|\mathbf{\Sigma}_{\mathcal{K}\mathcal{K}}\|_F \leq \|\mathbf{\Sigma}_{\mathcal{K}}\|_F.$$

■

C. Proof of the submodularity of $\|\mathbf{\Sigma}_{\mathcal{K}}\|_F$

Proof: Denoting set function $F(\mathcal{K}) = \|\mathbf{\Sigma}_{\mathcal{K}}\|_F$, then for all subsets $\mathcal{A} \subset \mathcal{B} \subseteq \mathcal{N}$ and $c \in \mathcal{N} \backslash \mathcal{B}$, we have

$$\begin{aligned} & F(\{\mathcal{A}, c\}) - F(\mathcal{A}) - (F(\{\mathcal{B}, c\}) - F(\mathcal{B})) \\ =& \left\|\mathbf{\Sigma}_{\{\mathcal{A},c\}}\right\|_F - \|\mathbf{\Sigma}_{\mathcal{A}}\|_F - \left(\left\|\mathbf{\Sigma}_{\{\mathcal{B},c\}}\right\|_F - \|\mathbf{\Sigma}_{\mathcal{B}}\|_F\right) \\ =& \frac{\|\mathbf{\Sigma}_c\|_F^2}{\|\mathbf{\Sigma}_{\mathcal{A}}\|_F + \left\|\mathbf{\Sigma}_{\{\mathcal{A},c\}}\right\|_F} - \frac{\|\mathbf{\Sigma}_c\|_F^2}{\|\mathbf{\Sigma}_{\mathcal{B}}\|_F + \left\|\mathbf{\Sigma}_{\{\mathcal{B},c\}}\right\|_F} \geq 0, \end{aligned}$$

hence $F(\mathcal{K}) = \|\mathbf{\Sigma}_{\mathcal{K}}\|_F$ is submodular. ■